$$= \frac{\left\{\begin{matrix}3300+5000+4000+4200+3500+4500+3200\\+3200+4100+4000+4300+3000+3200+4500+4100\end{matrix}\right\}}{15}$$

$$= \frac{58100}{15} = 3873.3$$

Can the amount of ₹3873.3 be given to all of them irrespective of their salary? Is ₹3873.3 is the suitable representative here? No, this is not suitable here because a student with family income ₹3000 and a student with family income ₹5000 will receive the same amount. Because the representative measure used here is not sutable for the above data, let us find the mode for this data.

3000		1
3200		3
3300		1
3500		1
4000		2

4100		2
4200		1
4300		1
4500		2
5000		1

Here mode is 3200 which means there are more number of students with a family income of ₹3200. But this does not suite our purpose.

Hence, mode is also not suitable. Is there any other representative measures that can be used here? Yes.

Let us look at another representative value which divides the data into two halves exactly. First, let us arrange the data in ascending order.

That is, ₹3000, ₹3200, ₹3200, ₹3200, ₹3300, ₹3500, ₹4000, ₹4000, ₹4100, ₹4100, ₹4200, ₹4300, ₹4500, ₹4500, ₹5000.

After arranging the income in ascending order, Rajam finds 8^{th} value (₹4000) which divides the data into two halves. It helps her to decide the amount of financial support that can be given to each of the students. Note that ₹4000 is the middle most value.

This kind of representative value which is obtained by choosing the middle item is known as Median.

Thus in a given data, arranged in ascending or descending order, the median gives us the middle value.

Consider another example, where the data contains even number of terms 13, 14, 15, 16, 17 and 18. How to find the middle term for this example? Here the number of terms is 6, that is an even number. So we get, two middle terms namely 3^{rd} and 4^{th} term. Then, we take the average of the two terms (3^{rd} and 4^{th} term) and the value we get is the median.

That is, Median $= \frac{1}{2}\left\{3^{rd} \text{ term} + 4^{th} \text{ term}\right\}$

$= \frac{1}{2}\left\{15 + 16\right\}$

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